

```
In [1]: #Creating conditional loops
for i in range(1,4):
    print(i*i,end=" ")
    print(i)
```

```
1 1
4 2
9 3
```

```
In [2]: #Create square root of numbers from 1 to 5
import math
for i in range(1,6):
    print((float)(math.sqrt(i)))
```

```
1.0
1.4142135623730951
1.7320508075688772
2.0
2.23606797749979
```

```
In [3]: #Print all the even numbers till 50
for i in range(1,51):
    if (i%2==0):
        print(i,end=" ")
```

```
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 4
8 50
```

```
In [4]: # Finding the factorial
n = 6
fact = 1
for i in range(1, n + 1):
    fact *= i
    print("The factorial of ",n," is: ",fact)
```

```
The factorial of 6 is: 1
The factorial of 6 is: 2
The factorial of 6 is: 6
The factorial of 6 is: 24
The factorial of 6 is: 120
The factorial of 6 is: 720
```

```
In [5]: #Number pyramid
for i in range(1,6):
    for j in range(1,i+1):
        print(j,end=" ")
    print()
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
In [6]: #First 20 natural numbers
for i in range(1,21):
    print(i,end=" ")
```

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

```
In [7]: #Fibonacci nos generation
f1, f2 = 0, 1
n = int(input("Enter the number: "))
for i in range(n):
    print(f1, end=" ")
    f1, f2 = f2, f1 + f2
```

Enter the number: 5
0 1 1 2 3

```
In [8]: #Fibonacci nos generation using lists
fibo=[0,1]
n=11
for i in range(2,n):
    fibo.append(fibo[-1]+fibo[-2])
    print(fibo)
```

[0, 1, 1]
[0, 1, 1, 2]
[0, 1, 1, 2, 3]
[0, 1, 1, 2, 3, 5]
[0, 1, 1, 2, 3, 5, 8]
[0, 1, 1, 2, 3, 5, 8, 13]
[0, 1, 1, 2, 3, 5, 8, 13, 21]
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55]

```
In [9]: #Sum of all even numbers from 1 to 50
sum=0
for i in range(1,51):
    sum+=i
    print(sum,end=" ")
```

1 3 6 10 15 21 28 36 45 55 66 78 91 105 120 136 153 171 190 210 231
253 276 300 325 351 378 406 435 465 496 528 561 595 630 666 703
741 780 820 861 903 946 990 1035 1081 1128 1176 1225 1275

```
In [10]: #Using while loop for to get fibonnaci
fibonacci=[0,1]
n=11
counter=2
while(counter<n):
    fibonacci.append(fibonacci[-1]+fibonacci[-2])
    print(fibonacci)
    counter+=1
```

```
[0, 1, 1]
[0, 1, 1, 2]
[0, 1, 1, 2, 3]
[0, 1, 1, 2, 3, 5]
[0, 1, 1, 2, 3, 5, 8]
[0, 1, 1, 2, 3, 5, 8, 13]
[0, 1, 1, 2, 3, 5, 8, 13, 21]
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55]
```

```
In [11]: #Print the vowels in the word
word = input("Enter the string: ")
arr_list = list(word)
vowels = []
for i in range(len(arr_list)):
    if arr_list[i].lower() in ['a', 'e', 'i', 'o', 'u']:
        vowels.append(arr_list[i])
print("Vowels in the string:", vowels)
print("Total number of vowels:", len(vowels))
```

```
Enter the string: Amulya
Vowels in the string: ['A', 'u', 'a']
Total number of vowels: 3
```

```
In [12]: #Reverse the string
value = input("Enter the element: ")
reversed_value = value[::-1]
print(reversed_value)
```

```
Enter the element: Amulya
aylumA
```

```
In [13]: #Print the characters of the string
name=str(input("Enter the name: "))
name_in_list=list(name)
for i in range(len(name)):
    print(name_in_list[i],end=" ")
```

```
Enter the name: Amulya
A m u l y a
```

```
In [14]: # The star pattern
for i in range(1,6):
    for j in range(1,i+1):
        print("*",end=" ")
    print()
```

```
*
**
***
****
*****
```

```
In [15]: # Factorial function in use
def factorial(n):
    fact = 1
    for i in range(1, n + 1):
        fact *= i
    print("The factorial of ",n," is: ",fact)
factorial(int(input("Enter the number:")))
```

```
Enter the number:15
The factorial of 15 is: 1
The factorial of 15 is: 2
The factorial of 15 is: 6
The factorial of 15 is: 24
The factorial of 15 is: 120
The factorial of 15 is: 720
The factorial of 15 is: 5040
The factorial of 15 is: 40320
The factorial of 15 is: 362880
The factorial of 15 is: 3628800
The factorial of 15 is: 39916800
The factorial of 15 is: 479001600
The factorial of 15 is: 6227020800
The factorial of 15 is: 87178291200
The factorial of 15 is: 1307674368000
```

```
In [16]: #Print square of numbers from 1 to 10
for i in range(1,11):
    print(i*i,end=" ")
```

```
1 4 9 16 25 36 49 64 81 100
```

```
In [17]: #Print the Prime numbers
import math
def is_prime(n):
    if n <= 1:
        return False
    for j in range(2, int(math.sqrt(n)) + 1):
        if n % j == 0:
            return False
    return True

number = int(input("Enter the number: "))
for i in range(2, number + 1):
    if is_prime(i):
        print(i, "is a prime number")
```

```
Enter the number: 10
2 is a prime number
3 is a prime number
5 is a prime number
7 is a prime number
```

```
In [18]: #Get the GCD of the numbers
def gcd(a, b):
    while b != 0:
        a, b = b, a % b
    return a
print(gcd(48, 18))
```

```
6
```

```
In [19]: # Importing the numpy module
import numpy as np
```

```
In [20]: # Using the arange function
arr=np.arange(1,10,2)
print(arr)
```

```
[1 3 5 7 9]
```

```
In [21]: # Using the array function
value=np.array(['a',1])
print(value)
```

```
['a' '1']
```

```
In [22]: #np fundamentals
data=np.array([[1,2],[3,4]])
print("The dimension is: ",data.ndim)
print("The cross dimension is: ",data.shape)
zero=np.zeros(4)
one=np.ones(4)
print(zero,end=" ", " ")
print(one)
np.empty((2,1))
```

```
The dimension is: 2
The cross dimension is: (2, 2)
[0. 0. 0. 0.], [1. 1. 1. 1.]
```

```
Out[22]: array([[ -2.00000000e+000],
               [ 3.9988166e-317]])
```

```
In [23]: # Bitwise operation handlings
x, y = 5, 6
result = x | y
print("Bitwise OR result:", result)
print("Binary result:", bin(result))
```

```
Bitwise OR result: 7
Binary result: 0b111
```

```
In [24]: # numpy array creation and working
a=np.arange(1,10)
b=np.linspace(1,10,5)
print("The array is: ",a)
print("The equal array is: ",b)
```

```
The array is: [1 2 3 4 5 6 7 8 9]
The equal array is: [ 1.   3.25  5.5   7.75 10. ]
```

```
In [25]: # Numpy functions
a=np.arange(12)
print(a.reshape(3,4))
print(a.ravel())
```

```
[[ 0  1  2  3]
 [ 4  5  6  7]
 [ 8  9 10 11]]
[ 0  1  2  3  4  5  6  7  8  9 10 11]
```

```
In [26]: # Numpy Operations
a=np.array([1,2,3,4,5])
b=np.array([6,7,8,9,10])
print("The add answer is: ",(a+b))
print("The sub answer is: ",(b-a))
print("The mul answer is: ",(a*b))
print("The div answer is: ",(b/a))
print("The pow answer is: ",(a**b))
print("The cop answer is: ",(a==b))
```

```
The add answer is: [ 7  9 11 13 15]
The sub answer is: [5 5 5 5 5]
The mul answer is: [ 6 14 24 36 50]
The div answer is: [6.         3.5         2.66666667 2.25
2.         ]
The pow answer is: [      1      128     6561    262144   9765625]
The cop answer is: [False False False False False]
```

```
In [27]: # Question-1: Pass check
marks=float(input("Enter the marks: "))
if (marks>=90 and marks<=100):
    print("You got A grade")
elif (marks>=75 and marks<=89):
    print("You got B grade")
elif (marks>=50 and marks<=74):
    print("You got C grade")
else:
    print("You are fail")
```

```
Enter the marks: 95
You got A grade
```

```
In [28]: # Question-2: Use while loop to find the sum of the digits
a=int(input("Enter the number: "))
val=0
while a>0:
    val+=a%10
    a = a//10
print("The sum of the digits are:", val)
```

```
Enter the number: 456
The sum of the digits are: 15
```

```
In [29]: # Question-3: Define the prime function prime(n)
import math
def prime(n):
    if n <= 1:
        return False
    for j in range(2, int(math.sqrt(n)) + 1):
        if n % j == 0:
            return False
    return True

number = int(input("Enter the number: "))
for i in range(2, number + 1):
    if prime(i):
        print(i, "is a prime number")
```

```
Enter the number: 25
2 is a prime number
3 is a prime number
5 is a prime number
7 is a prime number
11 is a prime number
13 is a prime number
17 is a prime number
19 is a prime number
23 is a prime number
```

```
In [30]: # Question-4: Palindrome check
a = input("Enter the string to check: ")
b = a[::-1]
print("Is the string palindrome? ", a == b)
```

```
Enter the string to check: Mom
Is the string palindrome? False
```

```
In [31]: # Question-5: Creating a 3x3 matrix and perform the operations-mul,
# mul==inverse
import numpy as np
A = np.array([[1, 2, 3],[4, 5, 6],[7, 8, 9]])
B = np.array([[9, 8, 7],[6, 5, 4],[3, 2, 1]])
mul = np.dot(A, B)
print("The dot product/Multiply is:\n", mul)
dett = int(np.linalg.det(mul))
print("The determinant is:", dett)
inv = np.linalg.inv(mul)
if (dett==0):
    print("No inverse")
else:
    print("The reverse exists")
```

```
The dot product/Multiply is:
[[ 30  24  18]
 [ 84  69  54]
 [138 114  90]]
The determinant is: 0
No inverse
```